

Norway

SLEIPNER

Volve F

F-1

F-1

Plan: 15/9-F-1 North Upside Pilot DG2 revp1

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 North Upside Pilot DG2 revp1	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-1					
Well Position	+N/-S	3.17 m	Northing:	6,478,566.69 m	Latitude:	58° 26' 29.9072 N
	+E/-W	-3.53 m	Easting:	435,046.49 m	Longitude:	1° 53' 14.7075 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-1				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	02.11.2005	-2.54	71.62	50,284

Design	15/9-F-1 North Upside Pilot DG2 revp1				
Audit Notes:					
Version:	Phase:	PLAN	Tie On Depth:	210.70	
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	3.17	-3.53	351.72	

Survey Tool Program	Date	16.08.2016			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)	
237.60	358.50	15/9 F1 26" Gyro Scientific Modular (F-1)	NOT_useMWD gyro, Baker I	Baker Hughes' MWD gyro service (GyroTrak)	
378.00	737.40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools	
756.80	1,335.60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,387.10	2,587.00	15/9 F-1 17 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
2,653.20	210.70	15/9 F-1 8 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
210.70	450.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Gyro, old	Other Gyro Tools	
450.00	650.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	NOT_useMWD gyro, Baker I	Baker Hughes' MWD gyro service (GyroTrak)	
650.00	1,355.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,355.00	2,585.00	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
2,585.00	3,684.37	15/9-F-1 North Upside Pilot DG2 revp1 (F-	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	3.17	-3.53	0.00	0.000	0.000	0.000	

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Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 North Upside Pilot DG2 revp1	Database:	Production EDM P246N

Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
150.15	0.00	0.00	150.15	3.17	-3.53	0.00	0.000	0.000	0.000
160.45	0.00	0.00	160.45	3.17	-3.53	0.00	0.000	0.000	0.000
170.04	0.00	0.00	170.04	3.17	-3.53	0.00	0.000	0.000	0.000
179.79	0.00	0.00	179.79	3.17	-3.53	0.00	0.000	0.000	0.000
189.74	0.00	0.00	189.74	3.17	-3.53	0.00	0.000	0.000	0.000
198.50	0.00	0.00	198.50	3.17	-3.53	0.00	0.000	0.000	0.000
210.70	0.00	0.00	210.70	3.17	-3.53	0.00	0.000	0.000	0.000
240.00	0.00	0.00	240.00	3.17	-3.53	0.00	0.000	0.000	0.000
250.00	0.00	0.00	250.00	3.17	-3.53	0.00	0.000	0.000	0.000
270.00	0.67	260.00	270.00	3.14	-3.65	0.00	1.000	1.000	0.000
283.00	1.10	260.00	283.00	3.11	-3.85	-0.01	1.000	1.000	0.000
300.00	1.10	260.00	299.99	3.05	-4.17	-0.02	0.000	0.000	0.000
330.00	1.10	260.00	329.99	2.95	-4.73	-0.04	0.000	0.000	0.000
360.00	1.10	260.00	359.98	2.85	-5.30	-0.05	0.000	0.000	0.000
390.00	1.10	260.00	389.98	2.75	-5.87	-0.07	0.000	0.000	0.000
420.00	1.10	260.00	419.97	2.65	-6.44	-0.09	0.000	0.000	0.000
450.00	1.10	260.00	449.97	2.55	-7.00	-0.11	0.000	0.000	0.000
463.00	1.10	260.00	462.96	2.51	-7.25	-0.11	0.000	0.000	0.000
480.00	1.73	283.74	479.96	2.54	-7.66	-0.02	1.500	1.116	41.893
510.00	3.11	298.29	509.93	3.04	-8.82	0.63	1.500	1.380	14.546
540.00	4.57	303.76	539.86	4.09	-10.53	1.92	1.500	1.456	5.470
570.00	6.05	306.57	569.73	5.69	-12.79	3.83	1.500	1.478	2.817
600.00	7.53	308.28	599.52	7.85	-15.60	6.38	1.500	1.486	1.711
630.00	9.02	309.43	629.21	10.56	-18.96	9.55	1.500	1.491	1.149
649.61	10.00	310.00	648.55	12.64	-21.46	11.95	1.500	1.493	0.867
660.00	10.00	310.00	658.78	13.80	-22.84	13.30	0.000	0.000	0.000
690.00	10.00	310.00	688.33	17.14	-26.83	17.19	0.000	0.000	0.000
720.00	10.00	310.00	717.87	20.49	-30.82	21.08	0.000	0.000	0.000
750.00	10.00	310.00	747.42	23.84	-34.81	24.97	0.000	0.000	0.000
780.00	10.00	310.00	776.96	27.19	-38.80	28.86	0.000	0.000	0.000
810.00	10.00	310.00	806.50	30.54	-42.79	32.74	0.000	0.000	0.000
840.00	10.00	310.00	836.05	33.89	-46.78	36.63	0.000	0.000	0.000
870.00	10.00	310.00	865.59	37.24	-50.77	40.52	0.000	0.000	0.000
900.00	10.00	310.00	895.14	40.58	-54.76	44.41	0.000	0.000	0.000
930.00	10.00	310.00	924.68	43.93	-58.75	48.30	0.000	0.000	0.000
960.00	10.00	310.00	954.22	47.28	-62.74	52.19	0.000	0.000	0.000
990.00	10.00	310.00	983.77	50.63	-66.74	56.08	0.000	0.000	0.000
1,020.00	10.00	310.00	1,013.31	53.98	-70.73	59.96	0.000	0.000	0.000
1,050.00	10.00	310.00	1,042.86	57.33	-74.72	63.85	0.000	0.000	0.000
1,080.00	10.00	310.00	1,072.40	60.68	-78.71	67.74	0.000	0.000	0.000
1,110.00	10.00	310.00	1,101.95	64.02	-82.70	71.63	0.000	0.000	0.000
1,140.00	10.00	310.00	1,131.49	67.37	-86.69	75.52	0.000	0.000	0.000
1,170.00	10.00	310.00	1,161.03	70.72	-90.68	79.41	0.000	0.000	0.000
1,200.00	10.00	310.00	1,190.58	74.07	-94.67	83.30	0.000	0.000	0.000

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Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 North Upside Pilot DG2 revp1	Database:	Production EDM P246N

Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,230.00	10.00	310.00	1,220.12	77.42	-98.66	87.19	0.000	0.000	0.000
1,260.00	10.00	310.00	1,249.67	80.77	-102.65	91.07	0.000	0.000	0.000
1,290.00	10.00	310.00	1,279.21	84.12	-106.64	94.96	0.000	0.000	0.000
1,320.00	10.00	310.00	1,308.76	87.46	-110.63	98.85	0.000	0.000	0.000
1,350.00	10.00	310.00	1,338.30	90.81	-114.62	102.74	0.000	0.000	0.000
1,358.04	10.00	310.00	1,346.22	91.71	-115.69	103.78	0.000	0.000	0.000
1,380.00	9.73	320.58	1,367.86	94.37	-118.33	106.79	2.500	-0.372	14.453
1,410.00	9.90	335.25	1,397.42	98.67	-121.02	111.44	2.500	0.172	14.672
1,440.00	10.67	348.62	1,426.94	103.73	-122.65	116.68	2.500	0.767	13.368
1,470.00	11.91	359.72	1,456.37	109.55	-123.21	122.52	2.500	1.246	11.103
1,500.00	13.51	8.50	1,485.63	116.12	-122.71	128.94	2.500	1.594	8.778
1,530.00	15.34	15.34	1,514.69	123.41	-121.14	135.94	2.500	1.834	6.838
1,560.00	17.34	20.69	1,543.48	131.42	-118.51	143.48	2.500	1.998	5.355
1,590.00	19.45	24.95	1,571.94	140.13	-114.82	151.57	2.500	2.111	4.252
1,620.00	21.64	28.38	1,600.03	149.53	-110.09	160.19	2.500	2.191	3.435
1,650.00	23.89	31.21	1,627.70	159.59	-104.31	169.32	2.500	2.249	2.824
1,680.00	26.18	33.57	1,654.88	170.31	-97.50	178.94	2.500	2.292	2.360
1,710.00	28.51	35.57	1,681.52	181.65	-89.68	189.03	2.500	2.325	2.002
1,731.83	30.22	36.85	1,700.54	190.28	-83.35	196.66	2.500	2.347	1.756
1,740.00	30.22	36.85	1,707.61	193.57	-80.89	199.56	0.000	0.000	0.000
1,770.00	30.22	36.85	1,733.53	205.65	-71.83	210.21	0.000	0.000	0.000
1,800.00	30.22	36.85	1,759.46	217.73	-62.78	220.87	0.000	0.000	0.000
1,830.00	30.22	36.85	1,785.38	229.82	-53.73	231.52	0.000	0.000	0.000
1,860.00	30.22	36.85	1,811.30	241.90	-44.67	242.17	0.000	0.000	0.000
1,890.00	30.22	36.85	1,837.23	253.98	-35.62	252.82	0.000	0.000	0.000
1,920.00	30.22	36.85	1,863.15	266.06	-26.57	263.47	0.000	0.000	0.000
1,950.00	30.22	36.85	1,889.08	278.14	-17.51	274.12	0.000	0.000	0.000
1,980.00	30.22	36.85	1,915.00	290.22	-8.46	284.77	0.000	0.000	0.000
2,010.00	30.22	36.85	1,940.93	302.31	0.59	295.43	0.000	0.000	0.000
2,040.00	30.22	36.85	1,966.85	314.39	9.64	306.08	0.000	0.000	0.000
2,070.00	30.22	36.85	1,992.77	326.47	18.70	316.73	0.000	0.000	0.000
2,100.00	30.22	36.85	2,018.70	338.55	27.75	327.38	0.000	0.000	0.000
2,130.00	30.22	36.85	2,044.62	350.63	36.80	338.03	0.000	0.000	0.000
2,160.00	30.22	36.85	2,070.55	362.72	45.86	348.68	0.000	0.000	0.000
2,190.00	30.22	36.85	2,096.47	374.80	54.91	359.33	0.000	0.000	0.000
2,220.00	30.22	36.85	2,122.40	386.88	63.96	369.98	0.000	0.000	0.000
2,250.00	30.22	36.85	2,148.32	398.96	73.02	380.64	0.000	0.000	0.000
2,280.00	30.22	36.85	2,174.24	411.04	82.07	391.29	0.000	0.000	0.000
2,310.00	30.22	36.85	2,200.17	423.12	91.12	401.94	0.000	0.000	0.000
2,340.00	30.22	36.85	2,226.09	435.21	100.18	412.59	0.000	0.000	0.000
2,370.00	30.22	36.85	2,252.02	447.29	109.23	423.24	0.000	0.000	0.000
2,400.00	30.22	36.85	2,277.94	459.37	118.28	433.89	0.000	0.000	0.000
2,430.00	30.22	36.85	2,303.87	471.45	127.34	444.54	0.000	0.000	0.000

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Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
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Planned Survey									
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2,460.00	30.22	36.85	2,329.79	483.53	136.39	455.19	0.000	0.000	0.000
2,490.00	30.22	36.85	2,355.71	495.62	145.44	465.85	0.000	0.000	0.000
2,520.00	30.22	36.85	2,381.64	507.70	154.50	476.50	0.000	0.000	0.000
2,550.00	30.22	36.85	2,407.56	519.78	163.55	487.15	0.000	0.000	0.000
2,580.00	30.22	36.85	2,433.49	531.86	172.60	497.80	0.000	0.000	0.000
2,583.67	30.22	36.85	2,436.66	533.34	173.71	499.10	0.000	0.000	0.000
2,593.67	30.22	36.85	2,445.30	537.37	176.73	502.65	0.000	0.000	0.000
2,610.00	29.47	33.91	2,459.47	543.99	181.43	508.53	3.010	-1.360	-5.397
2,640.00	28.29	28.17	2,485.74	556.39	188.91	519.72	3.010	-1.180	-5.735
2,670.00	27.37	22.03	2,512.27	569.05	194.85	531.39	3.010	-0.923	-6.139
2,700.00	26.73	15.56	2,539.00	581.95	199.25	543.52	3.010	-0.639	-6.471
2,730.00	26.40	8.87	2,565.83	595.04	202.09	556.07	3.010	-0.334	-6.692
2,760.00	26.38	2.09	2,592.71	608.29	203.36	569.00	3.010	-0.017	-6.775
2,790.00	26.68	355.39	2,619.56	621.67	203.07	582.28	3.010	0.300	-6.708
2,820.00	27.29	348.89	2,646.30	635.13	201.20	595.87	3.010	0.607	-6.501
2,850.00	28.18	342.71	2,672.86	648.65	197.77	609.74	3.010	0.893	-6.179
2,880.00	29.34	336.93	2,699.16	662.18	192.78	623.85	3.010	1.154	-5.781
2,910.00	30.72	331.58	2,725.14	675.68	186.25	638.15	3.010	1.385	-5.344
2,940.00	32.31	326.68	2,750.72	689.12	178.20	652.61	3.010	1.586	-4.897
2,970.00	34.06	322.22	2,775.83	702.46	168.65	667.19	3.010	1.758	-4.466
3,000.00	35.97	318.16	2,800.40	715.67	157.62	681.85	3.010	1.906	-4.063
3,030.00	38.00	314.46	2,824.37	728.71	145.15	696.55	3.010	2.030	-3.695
3,060.00	40.14	311.09	2,847.66	741.53	131.27	711.24	3.010	2.136	-3.367
3,090.00	42.36	308.02	2,870.22	754.12	116.01	725.89	3.010	2.225	-3.076
3,120.00	44.66	305.20	2,891.98	766.42	99.43	740.46	3.010	2.300	-2.821
3,150.00	47.02	302.60	2,912.88	778.41	81.57	754.90	3.010	2.364	-2.598
3,154.29	47.37	302.24	2,915.79	780.10	78.91	756.95	3.010	2.397	-2.482
3,180.00	47.37	302.24	2,933.20	790.19	62.91	769.24	0.000	0.000	0.000
3,184.73	47.37	302.24	2,936.40	792.05	59.97	771.50	0.000	0.000	0.000
3,210.00	44.83	302.22	2,953.93	801.76	44.57	783.33	3.010	-3.010	-0.030
3,240.00	41.82	302.19	2,975.75	812.73	27.15	796.70	3.010	-3.010	-0.033
3,270.00	38.81	302.15	2,998.62	823.06	10.72	809.29	3.010	-3.010	-0.038
3,300.00	35.80	302.11	3,022.48	832.73	-4.68	821.07	3.010	-3.010	-0.043
3,330.00	32.79	302.06	3,047.26	841.71	-19.00	832.02	3.010	-3.010	-0.050
3,360.00	29.78	302.00	3,072.90	849.97	-32.20	842.10	3.010	-3.010	-0.058
3,390.00	26.77	301.93	3,099.31	857.49	-44.26	851.28	3.010	-3.010	-0.070
3,420.00	23.76	301.84	3,126.44	864.26	-55.13	859.54	3.010	-3.010	-0.087
3,447.54	21.00	301.74	3,151.90	869.78	-64.04	866.29	3.010	-3.010	-0.109
3,450.00	21.00	301.74	3,154.20	870.24	-64.79	866.86	0.000	0.000	0.000
3,480.00	21.00	301.74	3,182.21	875.90	-73.94	873.77	0.000	0.000	0.000
3,510.00	21.00	301.74	3,210.21	881.56	-83.08	880.69	0.000	0.000	0.000
3,540.00	21.00	301.74	3,238.22	887.21	-92.22	887.60	0.000	0.000	0.000
3,570.00	21.00	301.74	3,266.23	892.87	-101.37	894.51	0.000	0.000	0.000

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 North Upside Pilot DG2 revp1	Database:	Production EDM P246N

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,597.54	21.00	301.74	3,291.94	898.06	-109.76	900.86	0.000	0.000	0.000	
3,600.00	21.00	301.74	3,294.24	898.52	-110.51	901.43	0.000	0.000	0.000	
3,630.00	21.00	301.74	3,322.24	904.18	-119.65	908.34	0.000	0.000	0.000	
3,660.00	21.00	301.74	3,350.25	909.83	-128.80	915.26	0.000	0.000	0.000	
3,684.37	21.00	301.74	3,373.00	914.43	-136.22	920.87	0.000	0.000	0.000	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude	
- hit/miss target										
- Shape										
F-1 North Upside Pilot N	0.00	360.00	3,151.90	869.78	-64.04	6,479,433.00	434,986.00	58° 26' 57.8790 N	1° 53' 10.0935 E	
- plan hits target center										
- Point										

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)		
220.90	220.90	30"	30.000	36.000		
1,355.00	1,343.22	20"	20.000	26.000		
2,585.00	2,437.81	13 3/8"	13.375	17.500		

Checked By: _____ Approved By: _____ Date: _____